Iterative Orthogonalization

Finding Orthonormal Basis

The famous Gram-Schmidt process is used to produce an orthogonal basis from a given basis. It provides a constructive proof of the earlier claim that every vector space has an orthonormal basis.

Gram-Schmidt

ALGOR *Input:* *collection* $\boldsymbol{x}_1, \ldots, \boldsymbol{x}_k$ *of linearly independent vectors.*

Output: collection $\boldsymbol{y}_1, \ldots, \boldsymbol{y}_k$ *of orthogonal vectors that span the same space.*

Process: Generate vectors $\boldsymbol{y}_1$, $\boldsymbol{y}_2$, $\boldsymbol{y}_3$, *...* *by*

$$\boldsymbol{y}_i = \boldsymbol{x}_i - \sum_{j=1}^{i-1} \mathrm{proj}_{\boldsymbol{y}_j}(\boldsymbol{x}_i)$$

These vectors can then be normalized, if desired.

Example

Find orthonormal basis of span of the vectors

$$\boldsymbol{x}_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \end{bmatrix} \qquad \boldsymbol{x}_2 = \begin{bmatrix} 2 \\ 0 \\ -1 \\ 3 \end{bmatrix} \qquad \boldsymbol{x}_3 = \begin{bmatrix} 3 \\ 1 \\ 1 \\ -7 \end{bmatrix}$$

Answer is:

$$\boldsymbol{y}_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \end{bmatrix} \qquad \boldsymbol{y}_2 = \boldsymbol{x}_2 - \frac{2}{2}\boldsymbol{y}_1 = \begin{bmatrix} 1 \\ -1 \\ -1 \\ 3 \end{bmatrix} \qquad \boldsymbol{y}_3 = \boldsymbol{x}_3 - \frac{4}{2}\boldsymbol{y}_1 - \frac{-20}{12}\boldsymbol{y}_2 = \begin{bmatrix} 8/3 \\ -8/3 \\ -2/3 \\ -2 \end{bmatrix}$$

Example Continued

After normalization we have the vectors $\frac{1}{\sqrt{2}}(1,1,0,0)$**,** $\frac{1}{\sqrt{12}}(1,-1,-1,3)$**, and** $\frac{1}{\sqrt{42}}(4,-4,-1,-3)$

Summary

The Gram-Schmidt process produces an orthogonal basis. It considers the basis vectors in turn and for each, subtracts its projection onto the previous basis vectors. The resultant basis can be made orthonormal by normalization.